

You cannot speak and listen at the same time: a probabilistic model of turn-taking

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Abstract Turn-taking is a preverbal skill whose mastering constitutes an important precondition for many social interactions and joint actions. However, the cognitive mechanisms supporting turn-taking abilities are still poorly understood. Here, we propose a computational analysis of turn-taking in terms of two general mechanisms supporting joint actions: *action prediction* (e.g., recognizing the interlocutor's message and predicting the end of turn) and *signaling* (e.g., modifying one's own speech to make it more predictable and discriminable). We test the hypothesis that in a simulated conversational scenario dyads using these two mechanisms can recognize the utterances of their co-actors faster, which in turn permits them to give and take turns more efficiently. Furthermore, we discuss how turn-taking dynamics depend on the fact that agents cannot simultaneously use their internal models for both action (or messages) prediction and production, as these have different requirements—or, in other words,

they cannot speak and listen at the same time with the same level of accuracy. Our results provide a computational-level characterization of turn-taking in terms of cognitive mechanisms of action prediction and signaling that are shared across various interaction and joint action domains.

Keywords Turn-taking · Action prediction · Signaling · Dialogue · Joint action

1 Introduction

The mastery of taking and giving turns is a preverbal skill that constitutes an important precondition for many social interactions and joint actions [93]. Turn-taking is important for several social abilities, such as building something together, playing team sports, musical improvisation, and in conversational domains, where it has been extensively studied [14,57,66].

Recent research on conversational turn-taking focused on universals in the underlying pattern of response latency of ten languages, with differences between the languages in the average gap between turns being within a range of 250 ms from the cross-language mean [100]. This and other studies have been interpreted in terms of a universal core principle of turn-taking, which Schegloff [96] calls a norm of “minimal-gap minimal-overlap” and which includes (i) the avoidance of overlapping talk (i.e., preventing the interlocutors from speaking at the same time) and (ii) the minimization of silence between conversational turns (i.e., preventing the interlocutors from being silent at the same time). Other studies challenge this interpretation by arguing that gaps and overlaps are frequent in conversations, and they are not necessarily indicators of trouble in communication or errors [46–48,109].

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Despite these advancements, the cognitive mechanisms underlying turn-taking are incompletely known. To address this problem, here we start from the premise that turn-taking is a kind of *joint action* [97,98]. We thus propose a computational-level characterization of turn-taking that uses two key mechanisms whose role is well established in several domains of joint action [97,98] including more recently conversational domains [42,44,45,62,66,89]: *action prediction* and *signaling*. We discuss how in joint action (e.g., conversational) settings these mechanisms permit recognizing an observed action (or heard utterance) faster, which in turn permits to “free” resources, e.g., stop using an internal model for utterance prediction and use it for planning the next utterance instead. Below we introduce *action prediction* and *signaling* mechanisms that we successively use in our simulation experiments, where we specifically ask how they permit agents to give and take turns efficiently and accurately.

1.1 Action prediction and signaling mechanisms in joint action

Current joint action theories assume that humans use predictive mechanisms (e.g., internal models) to understand the actions performed by others and predict their timing, so as to better coordinate with them [97]. Following a “motor cognition” approach, action prediction and recognition are assumed to rely on a *motor simulation* process: An agent internally simulates the action performed by another agent (i.e., the motor act she uses to utter a word) by reenacting the same motor patterns as those used in her own action (e.g., speech) production. In turn, this process automatically provides access to the action goal (e.g., the heard word) as well as to its low-level details (e.g., its timing) [52].

From a computational perspective, the motor simulation process underlying action prediction and recognition uses two kinds of internal models: forward and inverse. *Forward models* express the causal (forward) relationship between actions and their sensory consequences (e.g., between the production of an utterance and the perception of a specific auditory stream). A forward model, therefore, can be used to predict how the system’s state changes in response to a given motor command. *Inverse models*, on the other hand, perform the opposite transformation determining the motor command required to achieve some desired outcome (e.g., what phono-articulatory pattern is necessary to produce a specific utterance). Inverse models, therefore, act as controllers producing the motor command to bring the system about a desired goal, given the current state. In individual motor control, paired forward/inverse models are used to provide an accurate estimation of the current motor state and context, or to compensate for large feedback delays from the peripheral sensory system [55]. The same mechanism can be

reused in social settings by allowing the observing agent to predict and infer actions performed by its peers in terms of the observer’s action repertoire [110]. In this perspective, each internal model pair competes to explain the observable effects of others’ actions (e.g., inferring which utterance could have been uttered by an agent that has produced the heard auditory stream). A closely related view is that of *inverse inference* in generative models, where the hidden states of the model (i.e., the goal-directed action; for instance, a given utterance that is heard) are inferred from the noisy observations and prior information [39,40]. How are internal models learned in the first place is still subject to active research in the community (but see [20] for a recent account on computational models of model learning that adopt an intentional stance to prune the space of possible models during learning).

Although important, action prediction mechanisms might not be sufficient to deal with most real-world interactions and conversations, especially when many utterances are similar (and the associated internal models have similar kinematic or dynamic properties) and when the observation model is not accurate (i.e., the variance of the associated probability density function is high). We previously argued that, to alleviate this problem, people engaged in social interactions and conversations widely use *signaling* strategies; for example, they exaggerate their movements or speech to facilitate the recognition process by other agents [84,85] (we do not consider here other forms of signaling that use different communication channels, e.g., waving the hand or arm). There are various demonstrations that producers modulate their speech (e.g., loudness, choice of words, speech rate) depending on various contextual factors (e.g., amount of noise, prior knowledge or uncertainty of the comprehender) to help the comprehender. A well-known case is the exaggeration of the vowels in child-directed speech (“motherese,” [63]). These modulations are not only used during teaching, but also when the comprehension is difficult, as should be evident to anybody finding himself speaking more loudly and over-articulating in noisy environments (the so-called “Lombard effect”). Signaling was also recently studied in collaborative joint actions [9,72,83,92,104,105], fingerspelling [53], and in large-scale social interactions such as the collective behavior of an orchestra [17].

To understand the functioning of signaling in a conversational scenario, suppose an agent chooses an utterance and wants to ensure that the interlocutor understands and predicts it correctly. The goal of signaling is to choose the best parameters of the production model of the selected word (as encoded in the inverse–forward models of Fig. 1) to maximize the difference (in terms of perceived features) from the non-selected words; this process can also be called a “dis-similation” [53]. In addition, signaling must preserve the information content conveyed by the execution of models (i.e., ensure that the selected word is correctly uttered) and

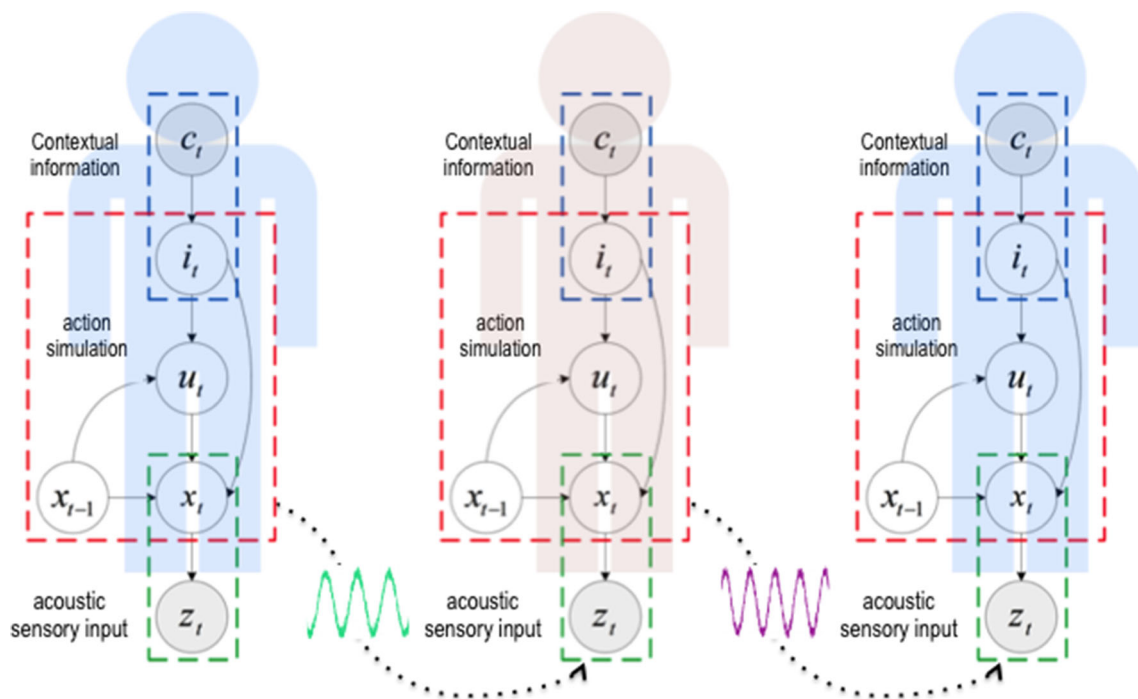


Fig. 1 A sample exchange between agent A and agent B. Here, agent A (*left*) utters a word (W2 in Fig. 2), which is heard by agent 2 (*center*). Agent 2, in turn, utters a word (W3 in Fig. 2), which is heard by agent A (*right*). The figure shows the control structure of agents A and B: a graphical model (DBN) based on coupled forward-inverse internal

models, whose variables are explained in the main text and in Table 1. The coupling between action and perception systems of agents A and B, respectively, is exemplified by the *black dotted edge* from node X (of agent A) to node Z (of agent B)

minimize the biomechanical cost of modifying the utterance (e.g., of exaggerations).

1.2 Action prediction and signaling in conversational turn-taking

We argue that the mechanisms of action prediction and signaling explained above underly also turn-taking abilities. If we consider a conversational turn-taking scenario, *action prediction* can refer to both the prediction (or recognition) of the last word of an utterance and the prediction of when the conversational turn of an agent will end (we will see later that these two predictions are closely related). Here, it is important to note that a heard message is a somewhat special kind of input: It is the result of an action performed by the speaker—hence the term *action* recognition. Advance information about the end of my turn allows you to start your turn at the right time, thus minimizing silence. In realistic dialogues, predictions are likely based on multiple elements: lexical, syntactic, semantic, and pragmatic [88], and the relative importance of these and other cues is disputed. Accordingly, there have been various proposals that prediction of end of conversational turns is based on prosodic elements (e.g., pitch and intonation) [56, 103] or lexico-syntactic elements [18].

A recent study suggests that the prediction of the end of turn depends on (or at least benefits from) the accurate anticipation of the last few words of a turn [69]. Coherently, in our approach the recognition of a message (or its last “word”) supports the prediction of the end of turn, as we explain below.

As we remarked, conversations are joint and collaborative actions and both speaker and listener contribute to its success. While listeners use action prediction mechanisms, speakers use signaling mechanisms. *Signaling* refers here to the intentional modulation of one’s own actions to make them more predictable; for example, during conversations a speaker can offer cues (e.g., modulate intonation or voice pitch) to signal that a message is about to end. In turn, this mechanism allows the listener to predict the message of the speaker faster and more accurately and to take a more timely turn. However, using action prediction and signaling might have shortcomings, too, because failing to predict (or signal) accurately might create overlaps between the turns (and the messages), which at least in some cases can hinder comprehension. An open question is thus whether action prediction and signaling mechanisms of the kind described in joint action domains support efficient turn-taking abilities in real-world scenarios and under conditions of uncertainty.

2 Methods: simulated conversational scenario

To answer this question, we take a computational approach and we compare the turn-taking performance of two agents with or without *action prediction* and *signaling* abilities. We simulate two agents, *agent A* and *agent B*, engaged in a simulated conversation where they interchangeably play the role of speaker and listener. The conversation is modeled as a sequence of exchanges. For each exchange (the minimum length conversation), one agent (say *agent A*) utters a word in a given dictionary (say $W_i \in \mathcal{D}$) having fixed and distinct length; then the other agent (say *agent B*) replies with another word (say $W_j \in \mathcal{D}$), and so on. Agents do not know beforehand which word will be uttered.

In our experimental setup, the objective of the two agents is to take turns efficaciously and accurately. This ability follows quite naturally from a widespread premise in motor cognition: That agents cannot (or at least have a limited ability to) use their internal models for both action prediction and performance simultaneously—because doing so would tap shared neuronal resources [52]. In a conversational scenario, agents cannot “speak and listen at the same time” with the same level of accuracy, because when an internal model is “occupied” (for example) in predicting an utterance, it cannot be simultaneously used for uttering another word, without generating interferences between the two tasks, in the same way simultaneously performing and observing two different actions would generate interferences [38, 58]. A second requirement for accurate turn-taking is that the agent recognizes the message and predicts the end of turn fast and accurately—and it is this ability that we test in our simulations.

Here, we focus on temporal and prosodic aspects of turn-taking, sidestepping for simplicity other aspects (e.g., semantic and pragmatic) of the interaction. Furthermore, in order to simplify the numerical comparisons between the experiments, we assume that an agent replies with a message

from its repertoire immediately after recognizing the (length of the) utterance of the interlocutor. However, as in real conversations, agents’ replies are not instantaneous, but rather have a *delay* (which is required to plan and initiate one’s own utterance). This is modeled here as a delay between the turns of the two actors. The delay depends on both the time that an agent needs to predict the end of turn, which is stochastic, and the time to plan her own reply. For the sake of simplicity, in our simulations this latter component has a fixed length for each agent; however, the model could accommodate variable delays depending on the complexity of the message to transmit. Finally, here we assume that the agent exchanges permutations of the same sequence of words. This is, however, not a necessity of the model but just a way to “equalize” the portions of dialogue of the two agents.

In our simulations, we test the trade-offs intrinsic in turn-taking and specifically in the necessity for an agent to recognize what has been uttered faster to “free” her internal models and plan her own response. An early recognition of the co-actor’s utterance gives a significant advantage as it allows starting the response earlier (and also, if required, avoiding “silence”). At the same time, an erroneous recognition implies that the agent would respond too early (which can also result in an “overlap”). How can this trade-off be efficiently managed?

3 Computational approach to action prediction and recognition

We implement action prediction and signaling mechanisms using the formalism of dynamic Bayesian networks (DBNs) [3, 73]. DBNs belong to the family of graphical models suitable for expressing conditional dependence between variables which ease the task of performing probabilistic inference. Figure 1 depicts our model in which—as usual—

Table 1 Variables and probability densities used in the graphical model of Fig. 1

(a) Variables			
c_t	Context (integer)	$[0, \dots, C]$	Prior over dictionary words
i_t	Utterance index (integer)	$[0, \dots, I]$	Words in the dictionary
u_t	Motor act (real)	ν (Hz), ϕ , L	Parameters specifying word production
x_t	State (real)	x	Acoustic signal, as shown in Fig. 2
z_t	Perceptual input (real)	x'	Acoustic signal as heard by the listener
(b) Probability densities			
$p(c_t)$	Contextual information		
$p(i_t c_t)$	Contextual induction		
$p(u_t x_{t-1}, i)$	Inverse model		
$p(x_t x_{t-1}, u_t, i)$	Forward model		
$p(z_t x_t)$	Observation model (prediction error)		

shaded nodes represent observed quantities while the others are hidden and need to be *estimated* through the process of probabilistic inference. Details of the model are given in Table 1.

In our case, we are interested in inferring and predicting the speaker's utterance i (e.g., word W_i) based on noisy perceptual (e.g., auditory) inputs z_t . Here, an utterance is treated like any other goal-directed action that is in both the speaker's and the listener's motor repertoire, and is associated with internal (inverse and forward) models for its execution [111]. Each word i_t is associated with a paired inverse–forward model, which permits production of the same word (e.g., a phono-articulatory internal model that permits uttering the word W_i). More specifically, each utterance i_t is associated with the generation of a motor control u_t , given the (hidden) state x_{t-1} (aka inverse model), and with the prediction of the next (hidden) state x_t , given the motor control u_t and the previous state x_{t-1} (aka forward model). Here, the states x intuitively correspond to parts of the uttered word; they are called “hidden” because they cannot be directly observed, but they can be inferred from the perceptual inputs z_t .

The red dashed square in Fig. 1 indicates that word recognition is a *motor simulation* process that reuses the same internal models as those used for the production of utterances. The most important difference between the two processes is that, during utterance recognition, the forward model predicts the word that the speaker is uttering. Essentially, the listener runs “in simulation” the motor commands associated with the most probable utterances in a given context. Through the forward models, they lead to different word-specific predictions for x_{t-1} . Then, the heard auditory input z_t is used to test which perceptual hypothesis was the most correct (i.e., produces less prediction error, see the green dashed square in Fig. 1), which in turn is an indicator of which utterance is being heard (here we adopt the maximum a posteriori decision rule). The blue dashed line indicates that the initial choice about which internal models to activate is biased by the a priori given contextual information, c_t (e.g., prior knowledge of a speaker's preference for some utterances), and the contextual induction process described through the probability distribution $p(i_t|c_t)$. While this feature has proven to be useful to prune huge hypothesis spaces, it will not be used in this work (i.e., we will use only one context, in which all models are equally likely to occur).

Given that in our model each utterance is encoded as a coupled forward–inverse model, predicting and understanding the actions performed by others correspond to computing the posterior distribution over possible forward–inverse action pairs, given all the observations so far, $p(i_t|z_{1:t})$. The theory of dynamic Bayesian networks provides a principled way to compute this latter quantity via a process called *filtering*. The general goal is to compute the probability distribution over

hidden (e.g., unobserved) variables of the model at time t , given the sequence of observed variables up to time t . Mathematically, let us denote with $\mathcal{X}_t$ the set of hidden variables at time t , and with $\mathcal{Z}_{1:t}$ the sequence of all available observed variables, the full conditional posterior ($p(\mathcal{X}_t|\mathcal{Z}_{1:t})$) can be obtained by the well-known recursive Bayesian inference scheme [73]:

$$\begin{aligned} p(\mathcal{X}_t|\mathcal{Z}_{1:t}) &= \eta p(\mathcal{Z}_t|\mathcal{X}_t) \cdot \int p(\mathcal{X}_t|\mathcal{X}_{t-1}) \cdot p(\mathcal{X}_{t-1}|\mathcal{Z}_{1:t-1}) d\mathcal{X}_{t-1} \end{aligned} \quad (1)$$

where $p(\mathcal{X}_t|\mathcal{X}_{t-1})$ and $p(\mathcal{Z}_t|\mathcal{X}_t)$ are called prediction and observation models, respectively, and η is the normalization factor that ensures that $p(\mathcal{X}_t|\mathcal{Z}_{1:t})$ is indeed a probability distribution (i.e., that sums to unity). In our model, these can be expressed as:

$$p(\mathcal{Z}_t|\mathcal{X}_t) = p(z_t|x_t) \quad (2)$$

$$p(\mathcal{X}_t|\mathcal{X}_{t-1}) = p(x_t|x_{t-1}, u_t, i_t) \cdot p(u_t|x_{t-1}, i_t) \quad (3)$$

$$p(\mathcal{X}_0) = p(x_0) \cdot p(c_0) \cdot p(i|c_0) \quad (4)$$

Finally, the particular distribution over possible forward–inverse action pairs we are interested in (i.e., $p(i_t|z_{1:t})$) can be obtained by the process of *marginalization* in which all other hidden variables are “summed out” from the full posterior distribution.

However, apart from special cases, the recursive propagation of the posterior density $p(\mathcal{X}_t|\mathcal{Z}_{1:t})$ above does not admit a closed form solution and approximate algorithms must be used instead. Here, we adopt *particle filters*, a Monte Carlo technique for sequential probabilistic inference [27]. Instead of analytically representing the required posterior distribution, we approximate it by a set of weighted samples (i.e., particles). In our setup, each particle represents a weighted hypothesis of an internal model activation in the utterance prediction task.

Following [85], signaling consists in a form of non-linguistic, sensorimotor communication, in which an agent tries to *dissimilate* his action from the alternative ones that are possible in the same context, while preserving its fundamental properties. For dissimulation, we intend the amplification of the differences between the selected action and the alternative one(s). Thus, for each given W_i , with $i \in \{1, \dots, I\}$, a sample drawn by its signaling distribution at time t should be representative of the distribution $p(\mathbf{x}_t|i_t = W_i)$ and at the same time should have a low probability of belonging to the other competing words, W_j , with $j \neq i$, $j \in \{1, \dots, I\}$.

To do this, a *rejection sampling*-based technique can be defined in order to obtain an approximation of the signaling distribution in a general case. Let $\hat{\mathbf{x}}_t$ be a sample from the dis-

tribution $p(\mathbf{x}_t|i_t = W_i)$. Given K random values, $u_k \in [0, 1]$ (one for each word), sampled from the uniform distribution over $[0, p(\mathbf{x}_t|i_t = W_k)/p_k^{\max}]$, we decide to *accept* the sample $\hat{\mathbf{x}}_t$ if the following holds:

$$u_i < b_i \cdot p(\hat{\mathbf{x}}_t|i_t = W_i) \quad \text{and} \quad u_j > b_j \cdot p(\hat{\mathbf{x}}_t|i_t = W_j), \\ \forall j \neq i \quad (5)$$

where $\mathbf{b} = [b_1, b_2, \dots, b_K]$ is a vector of weights that modulate the contribution of individual motor models to the signaling distribution. Thus, in the case of continuous distributions $p(\mathbf{x}_t|m_j)$, the formulation of the *signaling* distribution becomes:

$$p^{\text{sig}}(\mathbf{x}_t|W_i; \mathbf{b}) \\ \propto b_i \cdot p(\mathbf{x}_t|W_i) \prod_{j \neq i} \left(1 - b_j \cdot p(\mathbf{x}_t|W_j)/p_j^{\max}\right) \quad (6)$$

where $p_j^{\max}$ is the maximum value for the distribution $p(\mathbf{x}_t|m_j)$.

The weights $\mathbf{b}_i(t)$ can be computed minimizing the following:

$$\mathbf{b}_i(t) = \underset{\mathbf{b}(t)}{\text{argmin}} \left[KL \left[p_i^{\text{sig}}(\mathbf{b}(t)), p_i \right] + \lambda S \left(\theta - p_i^{\text{sim}} \right) \right] \quad (7)$$

where:

- $KL(\cdot, \cdot)$ is the Kullback–Leibler divergence between the signaling distribution with the set of weights $\mathbf{b}$ and the distribution with no signaling;
- λ is the amount of signaling in the given action;
- p_i^{sim} is an estimation of the agent’s posterior probability of correctly recognizing the word W_i (under the assumption that performer and perceiver share the same set of internal models);
- θ is the (experimental) threshold that the agent uses during word recognition;
- S is the logistic function.

The KL term considers the cost of signaling, where cost can be associated with biomechanical factors, effort, and other forms of costs (e.g., cognitive costs associated to planning and executing non-familiar or non-habitual movements). The λ term permits to modulate the amount of signaling ($\lambda = 0$ means no signaling, $\lambda = 1$ means max signaling). This minimization permits to estimate the potential benefit that the perceiver will have from signaling actions of having a different λ , where the benefit consists in better and faster recognition of the executed action.

This method for constructing a signaling distribution is meant to represent the trade-off that is intrinsic in it. Indeed,

Table 2 Parameters used in Eq. 8 for the generation of the preliminary dataset

Word name	ν (Hz)	ϕ	Note
W_1	8.18	0	$C - 1$
W_2	12.25	π	$G - 1$
W_3	16.35	0	$C0$
W_4	24.50	π	$G0$
W_5	32.70	0	$C1$
W_6	49.00	π	$G1$

as noted earlier, a signaling action has to simultaneously satisfy two constraints: It has to be easy to discriminate from the alternatives, and it has to be a valid action. For example, to dissimilate the vowel “a” from the vowel “e,” I can amplify some formants of the “a,” but not utter a “u.” The decision of how much to “deviate” from a standard “a” depends on a balance of the costs of deviations and the benefits for the listener—the latter is context sensitive as it depends on the listener’s uncertainty. These considerations are the rationale for the two terms in Eq. 7.

For simplicity, in the following simulations we assume that utterances are sampled from a dictionary composed of two to five words. However, the same method is general and can be used with an arbitrary number of internal models (and utterances). This approach has already shown promising results in real-world action prediction scenarios, such as in human action understanding [21], joint action [83], and robotics [19,76].

3.1 Simulated conversational scenario

To investigate this trade-off, we have performed multiple simulation experiments during which agents use (or do not use) action prediction and signaling strategies. To perform the simulation experiments, in a preliminary phase we first generated the internal models to be used for (simulated) speech production in our synthetic dictionaries composed of six “Words,” $W_1, W_2, W_3, W_4, W_5, W_6$. The words are generated starting from the sinusoidal waves of the following equation:

$$A \sin(2\pi \nu_i t + \phi_i) \quad (8)$$

The words share the same amplitude but different phases and frequencies, in order to reproduce frequencies of distinct musical notes (see Table 2). Figure 2 illustrates the synthetic words and Table 2 shows the parameters used for the generation of the preliminary dataset. The dataset is composed by 100 instances per word: Each word is generated according to the wave function of Eq. (8) with the addition of a Gaussian noise with standard deviation σ_{noise} in order to obtain a signal to noise ratio of $\text{SNR}_{\text{dB}} = 10 \log_{10}(\sigma_{\text{signal}}^2/\sigma_{\text{noise}}^2) \approx 17$ dB.

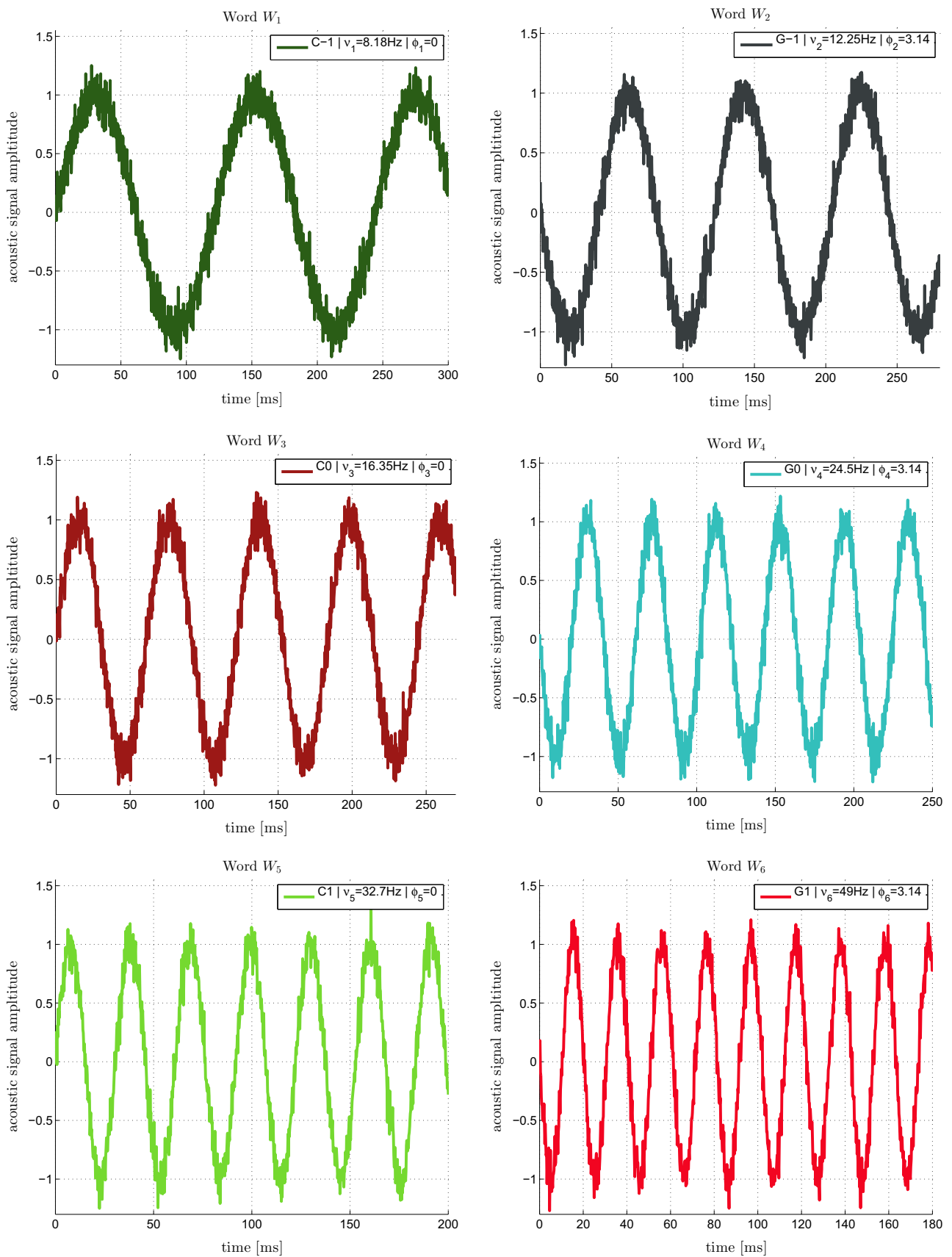


Fig. 2 Illustration of the synthetic words W_1 , W_2 , W_3 , W_4 , W_5 , W_6 used during simulations

Table 3 DBN modes and properties and scenario synthesis

Property	Mode	Property	Mode
	“Speaker” DBN (i clamped, x sampled)		“Listener” DBN (x clamped, i sampled)
S1	No signaling	L1	No recognition
S2	Signaling	L2	Recognition
Scenario		Agent A	Agent B
C1	Low cooperation: no recognition and no signaling	S1, L1	S1, L1
C2	Agent B recognition, no signaling	S1, L1	S1, L2
C3	Agent B recognition, agent A signaling	S2, L1	S1, L2
C4	Agent A, B recognition, no signaling	S1, L2	S1, L2
C5	Agent A, B recognition and signaling	S2, L2	S2, L2

We successively endowed the agents used in the simulations with (probabilistic) computational mechanisms for action prediction [21] and signaling abilities [85]. We modeled action prediction using the coupled forward/inverse modeling scheme [110], where the internal models used for action recognition are the same as those used for speech production (as introduced earlier). The models are cast as generative dynamic Bayesian networks and were obtained by regressing this synthetic data training set of 50 time series per word using a supervised learning technique, namely the Echo State Gaussian Process algorithm [11]. Here, the specific details of the implementation are not crucial (see [21]), but it suffices to say that the model takes the current sensory stimulus (i.e., the heard sound) as input and produces a probability distribution over the possible words that the speaker might have uttered; this probability distribution permits (i) the estimation of the most probable word from our dictionary and—by knowing the duration of each word—(ii) responding at the right time.

This pre-learning phase sets up the distributions constituting the DBN device simulating the agent. This DBN can be deployed into two ways (modes):

- S: A “Speaker” DBN mode that simulates an agent *producing* the acoustic signal. This configuration is obtained by *clamping* the variable i_t with the selected Word, and inferring the probability $p(x_t|i_{1:t})$, i.e., the distribution of the acoustic signal (constituting the output of this DBN mode) given the selected word;
- L: A “Listener” DBN mode that simulates an agent *recognizing* the acoustic signal. This configuration is obtained by *clamping* the variable z_t with the acoustic signal incoming from the co-actor (constituting the input of this DBN mode) and inferring the probability $p(i_t|z_{1:t})$, i.e., the probability of recognizing a word given the heard acoustic signal.

Moreover, each mode can be parametrized in order to express two properties for each mode. The “Speaker” mode can show:

- S1: No signaling property: The agent samples the acoustic from the distribution $p(x_t|i_t)$ of the model corresponding to the selected word W_i . This is accomplished by putting $\lambda = 0$ in the optimization process of Eq. (7);
- S2: Signaling property: The agent tries to dissimilate the spoken word that is obtained by sampling signaling distribution $p^{\text{sig}}(x_t|i_t = W_i)$ corresponding to the selected word W_i dissimilated with respect to the other possible words. This is accomplished by putting $\lambda = 1$ in the optimization process of Eq. (7).

The “Listener” mode can show:

- L1: No recognition property, i.e., the agent is not able to recognize the heard word, so it can only detect silence; this is achieved by setting the recognition threshold $\theta = 1$ (so as the agent is never confident of his recognition);
- L2: Recognition property, i.e., the agent tries to recognize the heard word in order to predict its end; this is achieved by setting the recognition threshold to $\theta < 1$ (in the simulations we choose $\theta \in \{0.65, 0.7, 0.75, 0.8, 0.85\}$) Note that this threshold refers to an internal parameter of the model, which determines how much evidence the model would need before being confident on the decision, not to a parameter of a statistical test (e.g., t test).

Table 3 sums up DBN modes and properties shown in the simulations and the resulting scenario that will be explained in the next session.

We recall that in the following simulations, an agent endowed with the signaling property (S2) uses the new “signaling” distributions for speech production. Figure 3 exemplifies the difference between the “original” and the “signaling” distributions of W1 versus W3. In Fig. 3(a), the two original distributions are shown, which overlap in several points (red areas), making them difficult to disambiguate for a listener. In Fig. 3(b), the two “signaling” distributions are shown: Here, the two distributions are modified compared to the original distributions so that they overlap to a significant

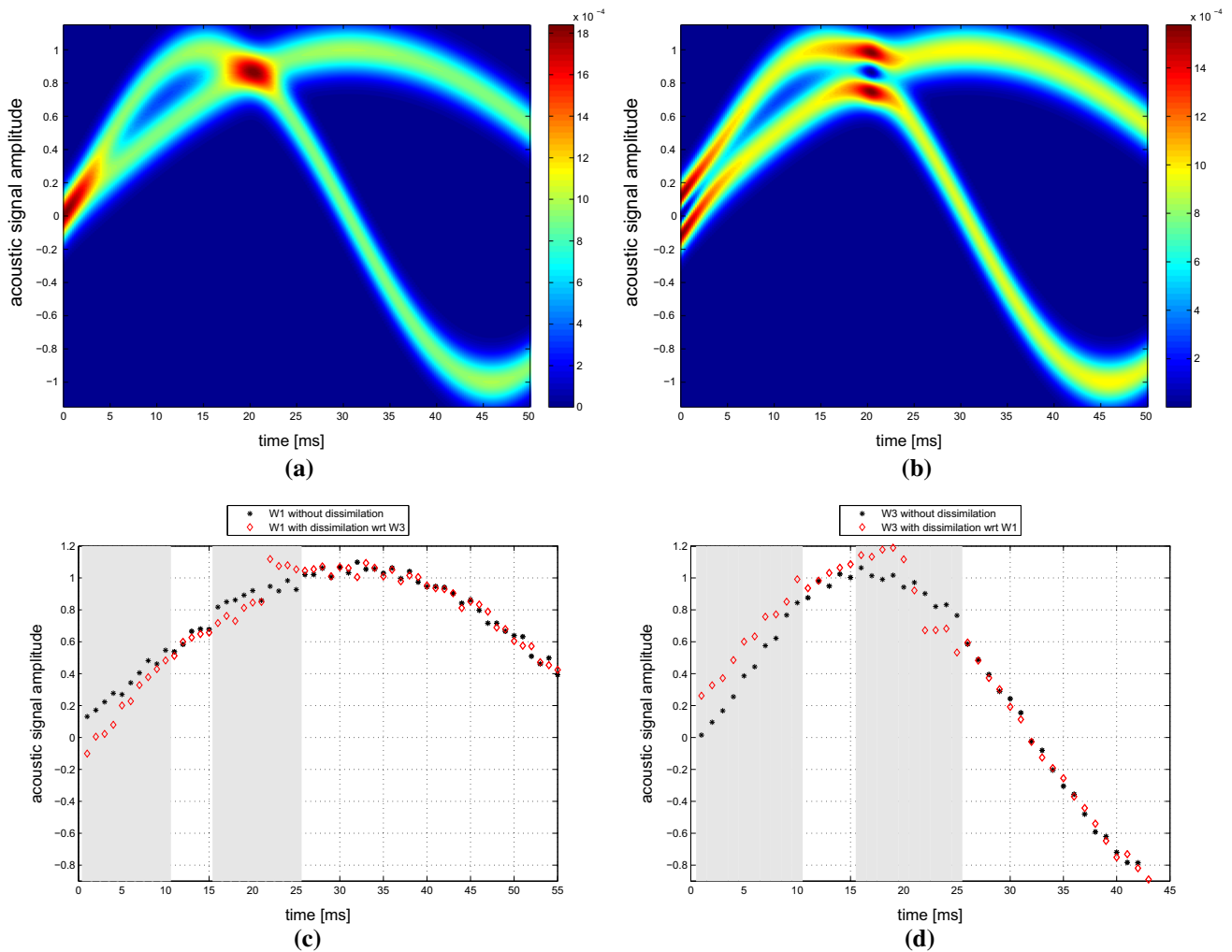


Fig. 3 Panels (a) and (b) show the values of the amplitude distribution of W1 and W3, i.e., a histogram of the probabilities at each time step t in colors from *blue* (zero) to *red* (high). Those distributions are obtained from the “Speaker” DBN equipped with property S1 (no signaling) and property S2 (signaling). The time sampling is 1 ms (abscissa axis), while the amplitude interval (ordinate axis) is divided in $\approx 10,000$ intervals, of size ≈ 0.23 in the interval $[-1.15, 1.15]$. The frequency of samples normalized per each time step gives the probability values in each bin, indicated in the colormap in the figure. In panel (a), the superposition of $p(x_t|i_t = W_1)$ plus $p(x_t|i_t = W_3)$ is shown when no signaling is present (S1). In panel (b), the corresponding superposition of proba-

bility densities $p^{\text{sig}}(x_t|i_t = W_1)$ and $p^{\text{sig}}(x_t|i_t = W_3)$ is presented in the signaling case (S2). It is possible to appreciate how the distributions “separate” to a greater extent in the signaling case. In panel (c), samples from $p(x_t|i_t = W_1)$ (in *black*)—distribution that refers to W_1 —versus samples $p^{\text{sig}}(x_t|i_t = W_1)$ (in *red*)—distribution that signals W_1 with respect to W_3 —are presented. It is possible to see how samples in the “ambiguous zones” depicted in gray are different in order to let the “Listener” DBN to faster recognize the correct word by integrating the contribution at each time step t . In panel (d), the symmetric case of $p(x_t|i_t = W_3)$ and $p^{\text{sig}}(x_t|i_t = W_3)$ is shown (color figure online)

lesser extent and can be better disambiguated. Note that signaling distributions do not correspond to different internal models; rather, they are created on the fly and are different depending on which the candidate words to be disambiguated are (e.g., the signaling distribution of W1 is different if the competing word(s) are W2, W3, or both).

4 Results

We have run five simulation experiments, each using one of five different dictionaries $\mathcal{D}_k = \{W_i\}_{i=1}^k$ of different sizes

(respectively, with 2, 3, 4, 5, 6 words). Each word has a different length. Each experiment includes 2000 exchanges: A set of 1000 words is uniformly drawn from the selected dictionary and each agent in a dyad utters a sequence of 1000 words, constituting a different permutation of this set of words. To compare the execution with different dictionaries, we set a different delay required to start one’s own turn (i.e., start uttering a word after recognizing the interlocutor’s word).

Before the actual simulation (i.e., exchange), we computed for each word W_i the *minimum response time* of the

model in Fig. 1 (when the probability crosses a fixed recognition threshold of 0.85) without signaling (r_i) and with signaling (r_i^S) as well as the *maximum response time* without signaling (R_i) and with signaling (R_i^S). The delay is set to $d_k = \min\{r_i, r_i^S\}_{i=1}^k$ where $k \in \{2, 3, 4, 5, 6\}$. The length L_i of each word W_i is chosen with the following criterion:

$$L_{i,k} = \max\{R_i, R_i^S\} + c \cdot 2d_k \quad (9)$$

with c being a random number uniformly sampled in $[0, 1]$. This mechanism that “scales” the word length—depending on both the minimum time necessary to recognize it and the dictionary size—permits a comparison of performance of the algorithm using dictionaries of different sizes.

We present five communication scenarios involving dialogues with different agents’ properties:

- C1. both agents use a baseline reactive strategy with no action prediction or signaling;
- C2. *agent B* uses action prediction;
- C3. *agent B* uses action prediction; *agent A* uses signaling;
- C4. both *agent A* and *agent B* use action prediction;
- C5. both *agent A* and *agent B* use action prediction and signaling.

Table 3 sums up how each scenario is set up in our framework, with each agent modeled by a DBN in different modes and with different properties.

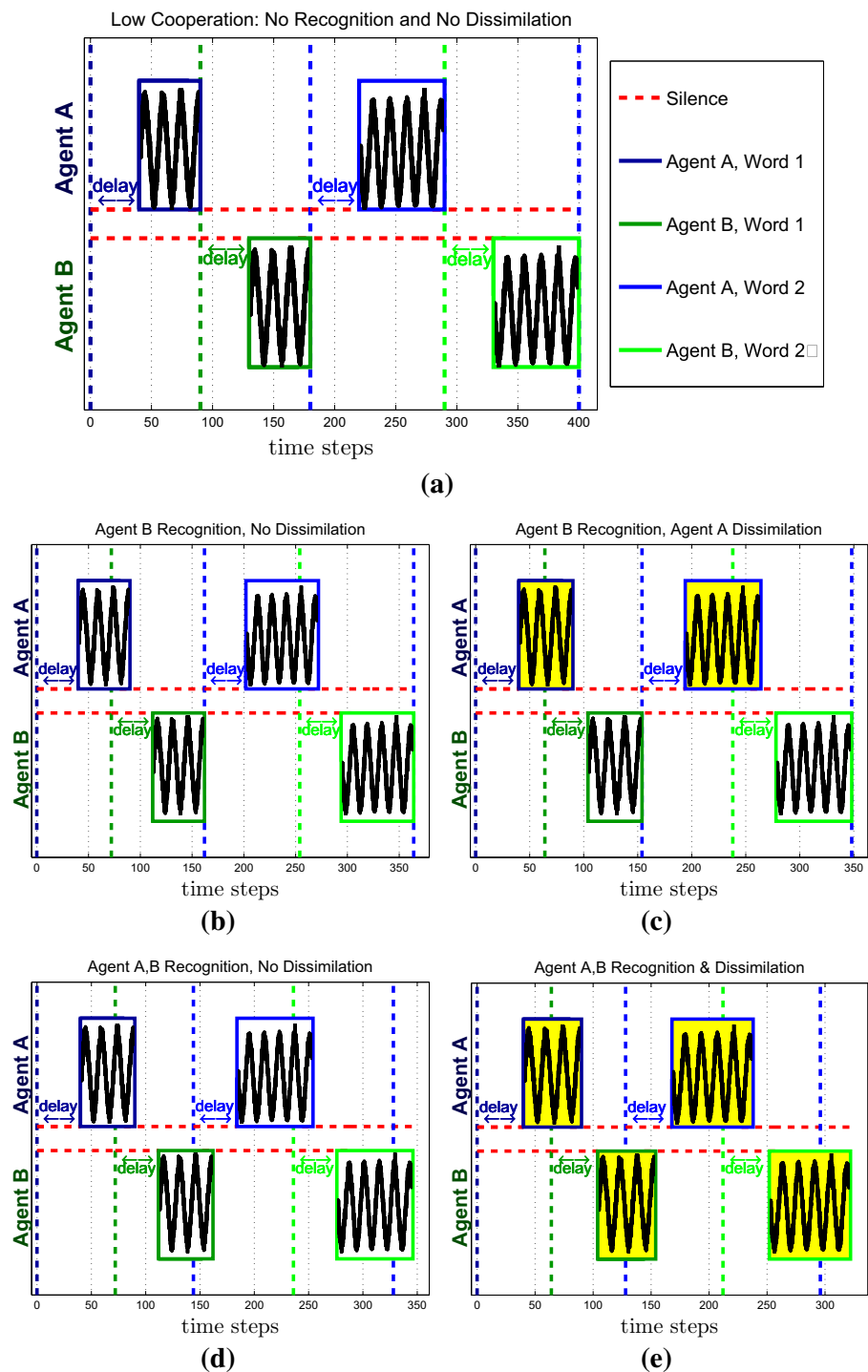
Figure 4 shows sample timelines of dialogues (here, for illustrative purposes, limited to only two exchanges in a parrot-like dialogue, i.e., each agent replies with the same word) with a dictionary of two words of different length ($\mathcal{D}_2 = \{W_1, W_2\}$) and it illustrates the differences between the scenarios. The blue vertical rectangles above the dashed horizontal lines represent the two words when they are uttered by *agent A*, while the green vertical rectangles below the dashed horizontal lines represent the same words when they are uttered by *agent B*. The dashed vertical lines indicate the moment when the word uttered by the speaker is recognized by the listener, while the red horizontal dashed lines indicate the time spent in silence. The behavior of the agents in the five scenarios is briefly explained in the following:

1. In the Baseline (reactive) scenario C1 (Fig. 4a), each agent waits until the end of the interlocutor’s word (see the vertical lines) and then utters its own word after a fixed delay. In this scenario, the possibility of overlapping utterances is excluded by design (as each agent waits for the end of the currently uttered word), at the expenses of the increased length of the overall dialogue which is evident from the long periods of silence (red horizontal dashed lines).

2. In the second communication scenario C2, *agent B* uses a more sophisticated strategy, *action prediction*, to reply proactively (Fig. 4b). That is, rather than just waiting for the end of the interlocutor’s message, it tries to estimate its length using action prediction (which can be more or less timely, as evident from inspecting the dashed vertical lines). In this way, it can reply proactively (i.e., start planning the reply before the completion of the interlocutor’s word), thus reducing the time spent in silence compared to the baseline condition. At the same time, as we will see below, this comes at the risk of augmenting the overlap between the messages (if the action prediction process was wrong).
3. In the third communication scenario C3, *agent B* uses action prediction mechanisms and at the same time *agent A* uses *signaling strategies* to facilitate *agent B*’s predictions (Fig. 4c). In this way, *agent B* can initiate her reply faster because (compared to the two preceding examples) it recognizes *agent A*’s utterances faster, thus further lowering the time spent in silence compared to the previous conditions.
4. In the fourth communication scenario C4, both *agent A* and *agent B* use action prediction but no signaling (Fig. 4d), further improving communication abilities with respect to the third scenario.
5. In the communication scenario C5, both *agent A* and *agent B* use action prediction and signaling (Fig. 4e), further improving communication abilities with respect to the fourth scenario.

As discussed earlier, we consider the ability to recognize an action, as fast and precisely as possible, an important requirement of joint actions. Such an ability permits to free up computational resources (e.g., an internal model occupied in recognizing an action) and—if required—to plan an appropriate response faster. To provide a formal comparison between the five communication scenarios outlined above—and to study the various trade-offs intrinsic in turn-taking, prediction, and signaling—we considered three different metrics: (i) total length of the dialogue, which—under the assumptions above—correlates with the ability to recognize words faster, (ii) time spent in silence, and (iii) time spent in overlap. We arranged the results of the comparison in Fig. 5 in a matrix of 5×3 sub-figures. The left column of Fig. 5 shows the results for the mean length of dialogue, the center column the time spent in silence, and the right column the time spent in overlap. The five figure rows correspond to the five dictionaries, from $\mathcal{D}_2$ (top) to $\mathcal{D}_6$ (bottom). Each sub-figure is a bar chart in which colored bars refer to scenarios (scenario C1 is dark green and scenario C5 is yellow). All the values are averaged over 10 executions, each composed of 2000 simulated exchanges. The values

Fig. 4 **a** Baseline (reactive) condition. In this and the following figures, an example of two exchanges is shown for the communication scenario C1, with $(\mathcal{D}_2 = \{W_1, W_2\})$. The blue vertical rectangles above the dashed horizontal lines represent the two words uttered by agent A, while the green vertical rectangles below the dashed horizontal lines represent the same words when uttered by agent B. Note that the two words have different length. The red dashed lines represent time steps spent in silence. The blue (green) vertical lines represent the instant when agent A (agent B) starts uttering a word; note that before the word is really uttered, there is a delay. **b** The same exchange as in (a), but now agent B uses action prediction, in this specific setup, word length prediction (communication scenario C2). **c** The same exchange as in (a, b), but now agent B uses action prediction and agent A uses signaling (communication scenario C3). **d** The same exchange as in (a–c), but now agents A and B use action prediction (communication scenario C4). **e** The same exchange as in (a–d), but now agents A and B use both action prediction and signaling (communication scenario C5)



are normalized with respect to the length of scenario C1 (baseline).

Each sub-figure has five sets of bars, corresponding to different choices of the “threshold” parameter of the action recognition model [21]. The parameter represents the minimum probability (ranging from 0.65 to 0.85) that algorithm has to assign to one of the hypotheses (in the case of a dictionary of two words, is the word I hear W_i or W_j ?) to

consider the action to be recognized. Intuitively, an agent using a threshold of 0.65 recognizes the action faster but this comes at the risk of a mis-recognition error, which in turn can increase overlaps. Conversely, an agent using a threshold of 0.85 is more confident about its prediction, but of course the prediction process takes longer and so part of the advantage in terms of faster response might be lost.

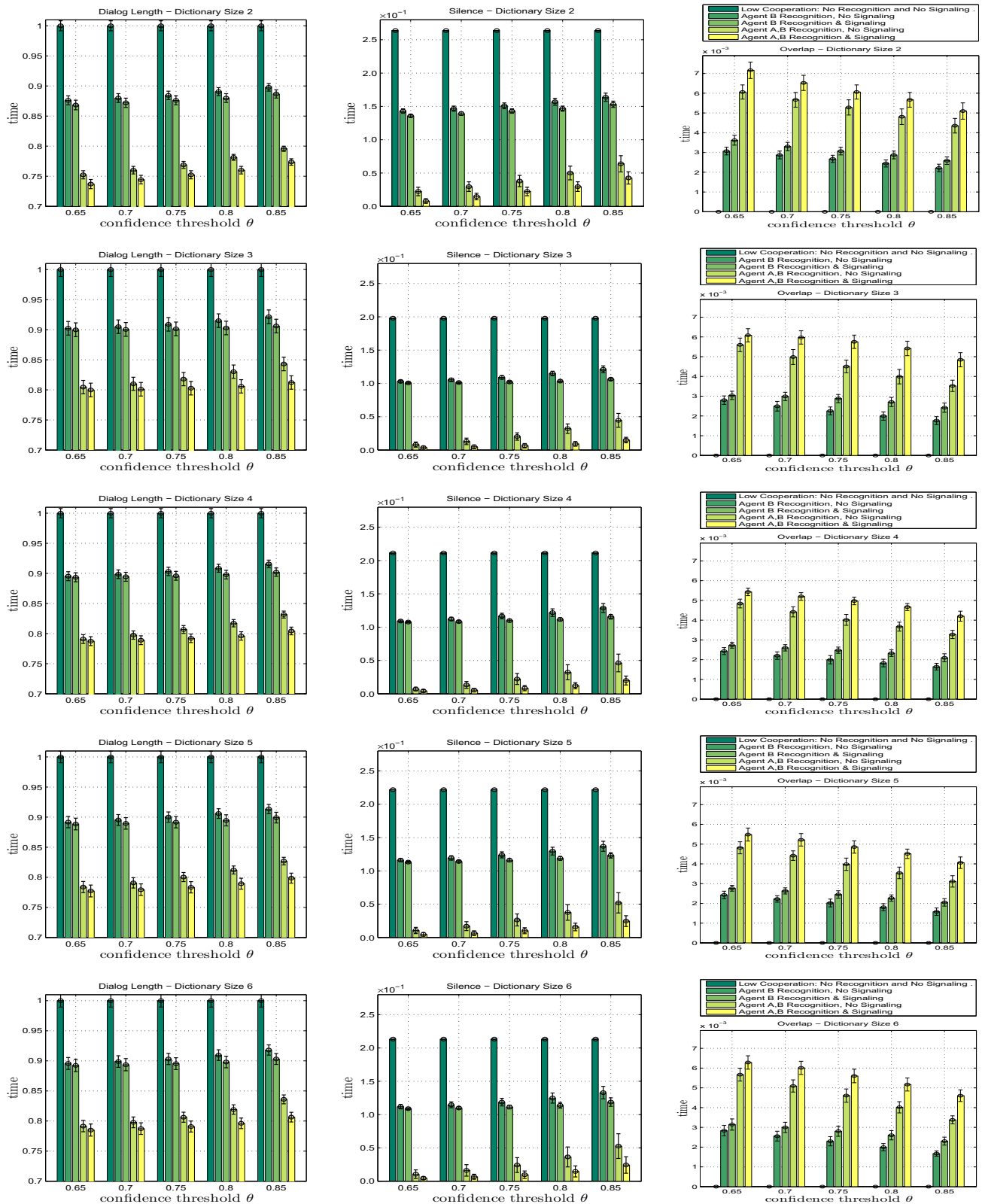


Fig. 5 Results of the simulations: mean and standard deviation of length of simulated dialogue (left column), silence during simulated dialogue (central column), and overlap time during simulated dialogue (right column). The five rows show the results for dictionaries of size

$\mathcal{D}_2$ (top row) to $\mathcal{D}_6$ (bottom row). Each colored bar corresponds to a scenario (see legend). Values are averaged over 10 executions, each composed of 2000 simulated exchanges, and normalized with respect to the length of scenario C1 (baseline)

The results of our experiments show that the overall decrease in interaction time (left column of Fig. 5) and the reduction in silence (central column) are very evident when passing from communication scenario C1 to scenario C5. The pattern of results is the opposite for overlaps (right column). Scenario C1 cannot have any overlap (by definition as its policy is to wait until the end of the utterance). Overlaps increase progressively when passing from C2 to C5—but note that the absolute size of overlap is minimal. As we were interested in quantifying the potential advantages of C5 compared to the other strategies, we performed a series of paired *t* tests to compare the dialogue length, silence and overlap of strategy C5 (using 0.85 as a threshold) against the other four strategies C1–C4, for all the dictionary sizes 2–6. This gives a total of 60 *t* tests: three parameters (dialogue length, silence, and overlap) by four comparisons (C5 vs. C1, C5 vs. C2, C5 vs. C3, and C5 vs. C4) by five dictionary sizes (2–6). All the *t* tests were significant ($p < 0.001$). This result confirms that using C5 brings significant advantages in terms of decreasing dialogue length and silence, while increasing overlaps. One may thus ask whether using both action prediction and signaling (C5) brings more benefits or costs to turn-taking and dialogue.

Answering this question in absolute terms would imply quantifying the benefits of minimizing dialogue time and the costs of overlaps—and both may be context sensitive (e.g., overlaps may be impolite in some social circumstances). However, it is instructive to compare the *absolute* changes in silence (central column of Fig. 5) and overlaps (right column of Fig. 5). This comparison shows that the values (and advantages) of silence are two orders of magnitude greater than the values (and disadvantages) of overlaps—or in other words, that decreases in silence are much larger than increases in overlap. This result suggests that, in absolute terms, action prediction and signaling abilities bring far more advantages than disadvantages. This can also be appreciated if one considers that, in principle, the overall length of a dialogue can be minimized by both reducing silence (which is positive) and increasing the overlap between the words of the two agents (which is negative). Our results show that reduction in silence is by far the most significant factor impacting the overall dialogue length. Finally, it is important to remark that listeners can modulate the amount of overlap when necessary, by setting a higher or lower threshold for recognition. This is already evident in Fig. 5, if one considers that overlaps decrease when the recognition threshold increases from 0.65 to 0.85. One can thus imagine that during dialogue, listeners may actively control these thresholds to trade off dialogue speed and potential overlaps in the ways they prefer (e.g., increase overlap when this is not socially unacceptable). In this perspective, using C5 is still more advantageous than the other strategies, since it extends the dynamical range that an agent can control (i.e., the quantity of silence and/or overlaps

he prefers), thus making dialogue more flexible and context sensitive. Flexibility produces a range of adaptive turn-taking strategies that are issued to specific context (e.g., engaging in a dialogue with a friend, a peer, or a master) and which can be potentially studied using this framework.

5 Discussion

Turn-taking has been widely studied empirically [14, 18, 44, 66] and computationally [50, 54, 61, 71, 91, 103, 108], but its underlying cognitive mechanisms remain incompletely known.

We have tested the hypothesis that two mechanisms that are well known to enhance coordination in joint actions, namely *action prediction* and *signaling*, might support efficient turn-taking, too [23, 85, 86, 97, 98, 105]. A premise of our argument is that the ability to give and take turns efficiently is a necessary requirement for agents that cannot (or are not fully able to) simultaneously use their internal models for two tasks that have competing requirements, action prediction, and production. Hence, the ability to understand an uttered message as fast as possible so as to “free” resources and plan an early response becomes important. This ability, in turn, permits to shorten or minimize (if needed) the length of dialogue. This perspective is not inconsistent with the idea of universal principles of avoidance of overlapping talk and minimization of silence between conversational turns [96, 100]. However, here these two imperatives are not imposed to the system but instead follow as consequences. Furthermore, our simulations offer a more nuanced view of the “minimum-gap, minimum-overlap” idea, as in fact even the more sophisticated strategies that we tested allow for some gaps and overlaps, as is indeed observed empirically [46–48, 109]. We consider these gaps and overlaps to be manifestations of the overall trade-offs that one must face in joint action and particular turn-taking domains such as conversations—where one can select among different modes, more reactive (but also slower) or more anticipatory strategies, depending on task demands.

The simulation experiments we performed aimed at comparing different strategies for turn-taking, ranging from a reactive strategy to more complex and cognitively plausible strategies that use action prediction and signaling. The pattern of results we obtained indicates that in simulated conversational setups both predictive and signaling strategies can be beneficial for turn-taking, especially if they are bidirectional. The ability to anticipate the end of another’s turn permits to “free” resources being used for the recognition, and to plan one’s own response faster, thus minimizing delays; however, it has the potential disadvantage of increasing overlaps (if prediction fails). In our simulations, the potential disadvantages of overlaps are minimal compared to the benefits.

The performance of the system remains stable with dictionaries of two to six words. The reason why the performance does not degrade is that we have used a scaling factor (Eq. 9) to set word length. This result indicates that the proposed system remains robust with an increased number of alternative words to predict and disambiguate. Indeed, at least in principle, the approximate Bayesian inference we used permits dealing with multiple internal models in parallel. However, when dealing with a very large number of words (and internal models) like in realistic conversational scenarios we expect the performance of the system to degrade. In these real-world situations, additional information becomes crucial to prune the search space; for instance, *prior* information on the production context can be added in the form of some prior hypothesis on the speaker's communicative intention [59]. In keeping with this view, during conversations humans use multiple sources of information and predict at different hierarchical levels in parallel (e.g., prosodic, syntactic, semantic), and higher-level information (e.g., about the meaning of a sentence) provides priors to the lower-level predictions (e.g., about the specific choice of words) and permits to contextualize them [79, 88]. Hierarchical extensions of our proposed Bayesian inference scheme that use multiple sources of information and multiple timescales could allow dealing with more realistic conversational scenarios and with multiple possible words and sentences [39, 83].

Taken together, the results of our simulation tests provide a proof of concept that the two mechanisms of *action prediction* and *signaling* increase the efficiency of turn-taking, especially when they are used in combination, as they permit inferring the uttered sentences better and faster and ultimately to give and take turns more efficiently. For the sake of simplicity, here we focused specifically on the temporal and prosodic aspects of turn-taking. Including in the model other important factors, such as conversational pragmatics and semantics, is an important objective for future research. For example, the semantics of the scene or the presence of certain objects in it might facilitate the prediction and recognition of the uttered sentences [33]. Including these factors in the model would plausibly require using *hierarchical* architectures, where some of the effects of semantics and pragmatics can deploy at longer time scales (because, e.g., objects remain in the scene for a long time) [34].

Our results support from a computational viewpoint recent theoretical models that suggest a role of action mutual prediction and the alignment of internal predictive models in dialogue, conversation as well as other joint actions [43, 98]. For example, in their recent proposal for a unified model of dialogue, Pickering and Garrod [89] have extensively discussed the importance of using mutual (motor) prediction to recognize another's actions and to monitor one's own actions. Friston and Frith [38, 41] implemented a computational model based on active inference (a scheme that uses

probabilistic inference) in which two agents (two simulated birds) align their internal generative models while singing, thus producing a coherent (and synchronized) flow of actions and predictions, and the concurrent execution (or “duet”) of the same song. The same system also reproduces turn-taking dynamics, by selectively raising and attenuating the precision of sensory information [41]. This system thus shows how to cast dialogue and turn-taking dynamics within a probabilistic (active inference) scheme that has mutual prediction at its core, but it does not incorporate signaling mechanisms that permit the performer agent to facilitate and guide the observer's expectations. The model proposed here is fully coherent with the prediction-based (probabilistic) inferential scheme used in these two computational studies and shows the importance of including signaling and sensorimotor communication mechanisms that are key to joint action dynamics including turn-taking [9, 83, 92, 104, 105].

The proposed model uses probabilistic (Bayesian) inference, in keeping with a number of recent studies applying the same principles across many domains of cognitive science, including action, perception, decision, and social interaction [8, 19, 21, 22, 24, 25, 36–38, 41, 60, 70, 78, 80, 81, 87, 101, 102, 110]. A key reason for using probabilistic schemes across all these domains is that they are all plagued by various sources of uncertainty, and Bayesian inference permits to handle uncertainty in rigorous ways. More broadly, recognizing the ubiquity of uncertainty has led to the proposal that the brain implements a uniform probabilistic inference scheme across all these domains, and other—or the Bayesian brain hypothesis [28]. This would have the additional benefit that one can appeal to a uniform computational scheme (and formal language) to explain various phenomena, rather than use a different framework for each of them. However, full Bayesian inference is computationally very costly and may even be intractable in realistic interactive domains. For this, most theories agree on the necessity of some form of approximation or “bounded rationality” but they differ on the nature of the approximation, with proposals ranging from variational inference [35, 77] to various forms of sampling including particle filtering [2, 7, 31, 95]. Interestingly, each specific approximation, variational or sampling-based, has specific functional and cognitive implications [94]. For example, sampling approaches such as the one used here converge to the optimal solution only in the limit of an infinite number of samples—although, in some cases, inference is accurate with few or even one sample [106]. At the cognitive level, one may imagine that the number of samples used for testing alternative hypotheses (e.g., during action observation) may correspond to a “limited resource” that varies as a function of initial uncertainty, importance of the task and/or cognitive load. It would be thus possible to test the specific quantitative predictions of our action observation method by manipulating (for example) the load and the number of par-

ticles available for the inference. Furthermore, by noticing that particle filtering re-weights samples (corresponding to different hypotheses) by the likelihood of the new observed data [26], one may manipulate the latter to test if the sampling method produces correct empirical predictions. Finally, it is worth noting that the rejection sampling method we use for constructing the signaling distribution has cognitive implication, too. In order to implement it, an agent has to maintain a representation of both the to-be-executed and the non-to-be-executed action—the latter is necessary to evaluate whether or not to reject a sample, see Eq. 5. This hypothesis is coherent with evidence that the brain actively maintains multiple motor plans in parallel, for example to select among them [12, 13, 82]. It remains to be assessed empirically if a parallel representation of motor plans can be used for planning signaling actions, too, as suggested by our computational scheme.

In sum, our simulative results show that not only prediction but also signaling mechanisms may improve conversational skills. Signaling refers here to an intentional effort to make one's own uttered words more discriminable and predictable by the interlocutor, alleviating the burden of real-time prediction and ultimately facilitating coordination [29, 83, 84, 92, 104, 105]. Even if in our case study prediction is rather easy and the size of dictionary limited, adopting a signaling strategy improves the reaction time of the listener, who can more easily recognize the communicative intentions of her interlocutor. Note that signaling can have more functions than those discussed here. For example, a speaker can use signaling to make the listener wait ("I'm not finished!"). Furthermore, a listener can use the feedback channel (e.g., nod) to signal that he has understood the sentence. Modeling these additional functions (and others that are revealed by recent turn-taking studies [44, 66]) would require extensions of this architecture, in which (for example) the speaker predicts when the listener will begin his conversational turn—and his possible recognition errors.

6 Conclusions

We have analyzed conversational turn-taking dynamics from a computational point of view, which focuses on the kind of problems that agents need to solve (e.g., decide when to start talking), the neuro-computational resources they can use for this (e.g., internal models for predicting and planning), and the strategies they can use (e.g., action prediction and signaling). Conversational turn-taking is an interesting case of joint action and its universality suggest it is part and parcel of the "interaction engine" of our (and other) species [65, 79]. From a computational perspective, negotiating turns efficiently (e.g., with little gaps and overlaps) is a challenging problem and requires agents to solve (implicitly or explicitly) a trade-off between the potential benefits of having

a shorter conversation with no gaps, and its costs such as the risk of overlaps and errors or misunderstanding. Human data showing that gaps in turns are typically shorter (about 200 ms [66]) than the time required to plan an utterance strongly suggest that predictive mechanisms are in play—or in other words, that listeners predict the speakers' end of turn based on (for example) syntactic and prosodic information and start planning their next sentence early on. Yet, action prediction/understanding can be challenging, especially with strict temporal constraints, and thus agents can benefit from a mixed strategy that uses action prediction in combination with signaling mechanisms. Our simulative results clearly show that it is the combination of (bidirectional) prediction and signaling mechanisms, which brings the best results. Indeed, taken together, these two mechanisms permit a more reliable representation of the input and more discriminable input patterns (i.e., a better separated input space), thus improving prediction and allowing faster action selection, with less error in choice of potentially conflicting actions. These computational results suggest that optimal turn-taking benefits from a combination of action prediction and signaling mechanisms; yet in practice, using these mechanisms may be costly, and thus their selection would depend on a form of cost-benefit optimization. For example, a speaker may stop signaling when s(he) knows the listener has correctly understood a sentence; in the same way, a demonstrator agent stops signaling when his or her co-actor has correctly understood a joint goal [64, 83].

Of note, here we assume that the shaping of the input space is done using an intentional signaling strategy. This view of signaling parallels theoretical considerations on the importance of sending "coordination signals" during joint actions and conversations [14]. Indeed, in the linguistic literature, our proposed signaling mechanisms resemble *recipient design* (or audience design) mechanisms: *Agent A* models *agent B*'s recognition process and facilitates it [14, 15], but see [6, 30] for different interpretations. A first element distinguishing our proposal from recipient design theories is that—we argue—signaling mechanisms could reuse the same internal models as those involved in action prediction perception and action systems. This view is coherent with a motor cognition approach, according to which shared neuronal resources underlie action execution, recognition, and simulation [52] (see [90] for a discussion of the status of motor theories of cognition in the domain of language processing) and with recent evidence showing that action prediction mechanisms can be reused for recipient design during communication [74, 75].

A second difference between our view and recipient design views is that the latter tend to assume that speakers have both the intention to communicate something and the intention to expose their own communicative intentions [14]. The latter intention is not strictly speaking necessary in the

framework we have proposed. However, exposing one's own intention (and conversely recognizing that another has a communicative intention) might play an important and additive role in pedagogical contexts [16], see also [99]. It remains to be established if these mechanisms give advantages in our simulated setup and more in general whether communicative intention recognition is a necessary ingredient of successful conversational joint actions.

Our proposal is also related to the *hyperarticulation and hypoarticulation* model [68], which suggests that speakers actively stress phonetic forms in ambiguous (noisy) situations or when words are not easily predictable. In keeping, various empirical studies support the idea that hyperarticulation increases as lexical predictability decreases [1,67]. Our proposal recognizes hyperarticulation as a valid mechanism for signaling and additionally suggests that speakers may use hyperarticulation (or more broadly, modulate phonetic forms) in different ways depending on the spoken word and its alternatives. In other words, one may signal the same word (say, house) differently if (the speaker believes that) the listener is uncertain between "house" and "mouse" or between "house" and "pause." Because signaling is a general mechanism that allows the listener to disambiguate the correct alternative better and faster, it has to be sensitive to the context in which the word is uttered (e.g., the competing hypotheses) and not just to noise levels.

The use of signaling has some interesting consequences on the way dialogues unfold in time. First, the demands for signaling are increased under various forms of uncertainty, which may be due to noise or the choice of unpredictable or low-frequency words. This would imply that maintaining the same efficiency of dialogue and turn-taking would require extra effort from both the listener and the speaker. Second, signaling has no benefits if the listener can correctly predict or recognize the speaker's sentence; hence, a speaker can stop signaling when the listener shows signs that he understood. A counterintuitive prediction would be that when two co-actors know each other's next sentence, as during a theater piece, they would (spontaneously) offer less coordination cues, hence the quality of the dialogue would lower (but of course professionals would have ways to overcome this problem).

Our results can also help shedding light on what are the cognitive and social/interactional prerequisites to acquire turn-taking abilities during development. Indeed, turn-taking abilities are not innate but rather children (and even infants) develop the ability to take and give turns through repeated interactions with their parents and other children [10]. The importance of action prediction for turn-taking suggests that the development of predictive abilities could constitute a key precondition for the mastery of turn-taking. It has been reported that the maturation of prediction skills precedes and supports the formation of motor skills during development [49] and learning [32]. Similarly, it is possible to specu-

late that during development predictive abilities constitute an important building block to improve turn-taking, shifting from reactive to anticipatory strategies of coordination.

Evidence that conversations can proceed with short gaps suggests the use of quite accurate predictive mechanisms [100]. We thus expect turn-taking abilities to become more efficient when prediction abilities fully mature. We suggest that in this critical developmental period teachers consistently exploit signaling strategies to scaffold the children's predictive abilities and ultimately their acquisition of turn-taking. Many examples of signaling during teaching have been documented, including the over-articulation of vowels in child-directed speech (motherese) [63], the over-articulation of gesture in infant-directed action (motionese) [4,5], and ostensive movements in pedagogical contexts [16]. These signaling dynamics may help establishing coherent dyadic interactions, such as for example vocal rhythm coordination between child and caregiver [51], which in turn would facilitate and scaffold learning and development. More broadly, efficient learning and development may rest on *social feedback loops* [107] and signaling may be an important mechanism to support these loops and keep them "in register." Signaling strategies might be used also afterward (i.e., when children's prediction abilities mature), but to a lesser extent unless specifically required by the situation (e.g., in noisy environments or novel situations). This hypothesis is consistent with evidence indicating that signaling strategies are less used if the interlocutors' uncertainty diminishes [83].

It is worth noting that if—as we have argued—signaling abilities reuse internal models involved in action prediction, teaching how to predict and teaching how to signal should go hand in hand. In sum, social interactive loops created by the teacher (and reinforced through children–teacher feedback loops) could lead to a progressive maturation of predictive, signaling, and turn-taking abilities during development. Because we consider predictive abilities to be important for sentence *understanding* and not only prediction (of end of turn), ultimately these interactional dynamics scaffold the development of efficient mechanisms for linguistic comprehension and the negotiation of meaning and joint actions through speech acts.

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